

Air Quality R data set analysis

1. Objective

The objective of this project is to analyze the factors that are influencing the air quality, focusing on the ozone levels.

For the analysis made in this project are used predictive models and statistical techniques (ANOVA, Nonparametric testing, Normality testing, Independent samples t-test, test for proportion, etc.) in order to understand how the meteorological variables that are present in the used data set affect the ozone. On the other hand, this project determines if there exists statistically significant differences in air quality between the months available in the data set.

2. Data set

The data set used in this project is called “airquality” R package which provides daily air quality measurements from New York from may to september 1973. This data set contains 153 rows and 6 columns. Those columns, or variables, are:

Ozone – ozone levels;

Solar.R – the solar radiation;

Wind – average wind speed;

Temp – maximum daily temperature;

Month – the month that the data is registered from.

3. Methodology

The data set “airquality” was loaded (Fig. 1).

```
#Loading the data  
data("airquality")
```

(Fig. 1 “Loading the data set”)

First of all it was checked if there exists missing values in the rows of the data set in order to clear them to not have problems in the next steps of the project (Fig. 2).

```
#Checking for missing values  
summary(airquality)  
  
#Removing rows with missing values  
aq_clean <- na.omit(airquality)
```

(Fig. 2 “Checking for missing values and deleting them”)

As shown in Fig. 3, there was identified 37 null values in column “Ozone” and 7 null values in column “Solar.R”. Those null values were removed from the data set to ensure the integrity of the results. The missing values could prevent the predictive model from learning accurate patterns and cause errors in the calculations of correlation matrix and other descriptive statistics. Also in Fig. 3 are presented details for the minimum and maximum values, median, mean and 1st and 3rd quartile of every column.

Ozone	Solar.R	wind	Temp	Month	Day
Min. : 1.00	Min. : 7.0	Min. : 1.700	Min. :56.00	Min. :5.000	Min. : 1.0
1st Qu.: 18.00	1st Qu.:115.8	1st Qu.: 7.400	1st Qu.:72.00	1st Qu.:6.000	1st Qu.: 8.0
Median : 31.50	Median :205.0	Median : 9.700	Median :79.00	Median :7.000	Median :16.0
Mean : 42.13	Mean :185.9	Mean : 9.958	Mean :77.88	Mean :6.993	Mean :15.8
3rd Qu.: 63.25	3rd Qu.:258.8	3rd Qu.:11.500	3rd Qu.:85.00	3rd Qu.:8.000	3rd Qu.:23.0
Max. :168.00	Max. :334.0	Max. :20.700	Max. :97.00	Max. :9.000	Max. :31.0
NA's :37	NA's :7				

(Fig. 3 “Number of null values identified”)

Another step before other calculations is to normalize the month into a factor. It does not make any sense to make the algorithm think of the month as a simple number, so the number values were transformed into labels (Fig. 4).

```
#Converting the month to a factor
aq_clean$Month <- as.factor(aq_clean$Month)
```

(Fig. 4 “Transforming the month variable from number to a simple label”)

The new table with the cleared and normalized data is presented in Fig. 5. There are no more null values and the month number is now considered as a label.

Ozone	Solar.R	wind	Temp	Month	Day
Min. : 1.0	Min. : 7.0	Min. : 2.30	Min. :57.00	5:24	Min. : 1.00
1st Qu.: 18.0	1st Qu.:113.5	1st Qu.: 7.40	1st Qu.:71.00	6: 9	1st Qu.: 9.00
Median : 31.0	Median :207.0	Median : 9.70	Median :79.00	7:26	Median :16.00
Mean : 42.1	Mean :184.8	Mean : 9.94	Mean :77.79	8:23	Mean :15.95
3rd Qu.: 62.0	3rd Qu.:255.5	3rd Qu.:11.50	3rd Qu.:84.50	9:29	3rd Qu.:22.50
Max. :168.0	Max. :334.0	Max. :20.70	Max. :97.00		Max. :31.00

(Fig. 5 “Cleared and normalized data set”)

A first step in analyzing this data set is to find out the standard deviation for the analyzed variables of this project, ozone and temperature (Fig. 6).

```
> #Standard deviation for ozone and temperature
> sd(aq_clean$Ozone)
[1] 33.27597
> sd(aq_clean$Temp)
[1] 9.529969
```

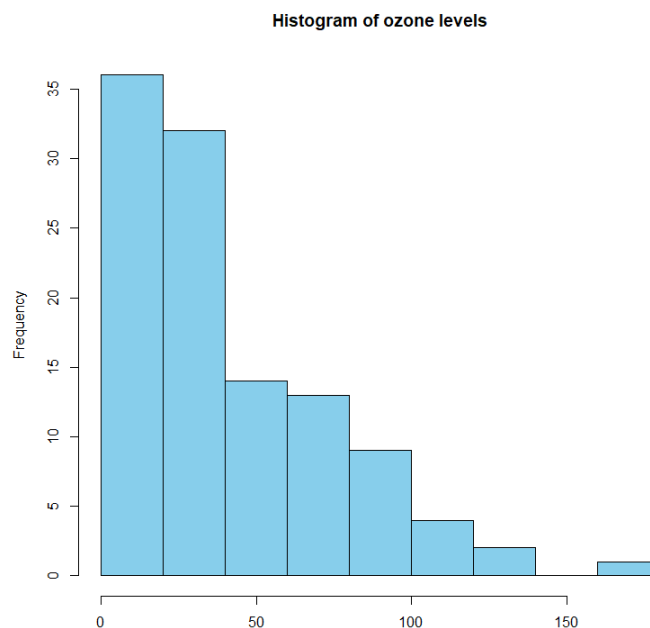
(Fig. 6 “Standard deviation for variables ozone and temperature”)

The standard deviation of the ozone levels value of approx. 33,28 presents that the values are not consistent and fluctuates a lot, so in some days the air pollution is small, in other days the pollution can be very strong. On the other hand, the standard deviation of temperature, which is around 9,53

is much better because it means that the temperature values does not fluctuate that much, so the temperatures in the studied period are relatively stable.

In order to be able to have more views on the ozone situation, there were created some visualizations. The first one is a histogram that presents how often different ranges of ozone levels occurred during the observation period (Fig. 7).

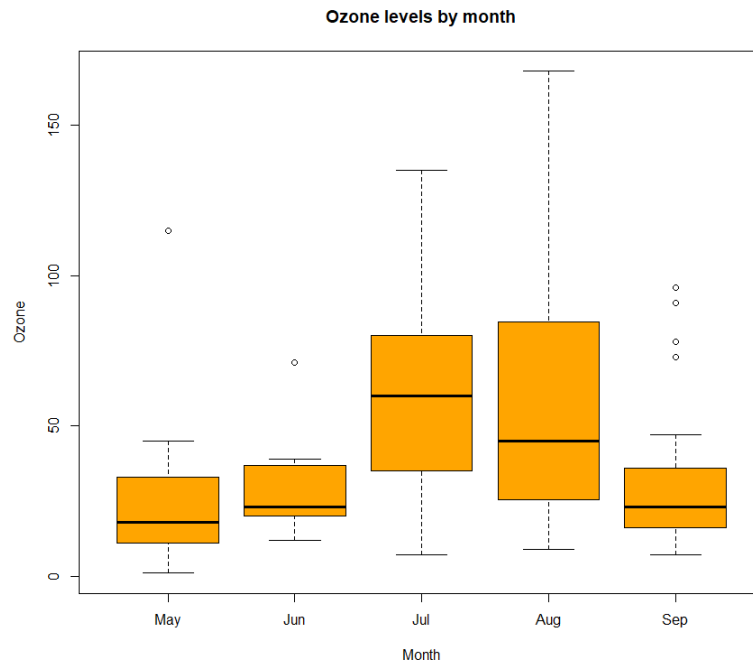
The ozone levels were over 35 times between 0-20 and a bit over 30 times between 20-40. Those values of the ozone levels, up to 40-50 are considered “safe”. In the other cases, the ozone level was at a “moderate” level maximum of 14 times (values between 50-100) and the values over 100, which are considered “risky”, were registered on the maximum of 5 times, with a very big value at the end between 160-180.



(Fig. 7 “Ozone levels frequency”)

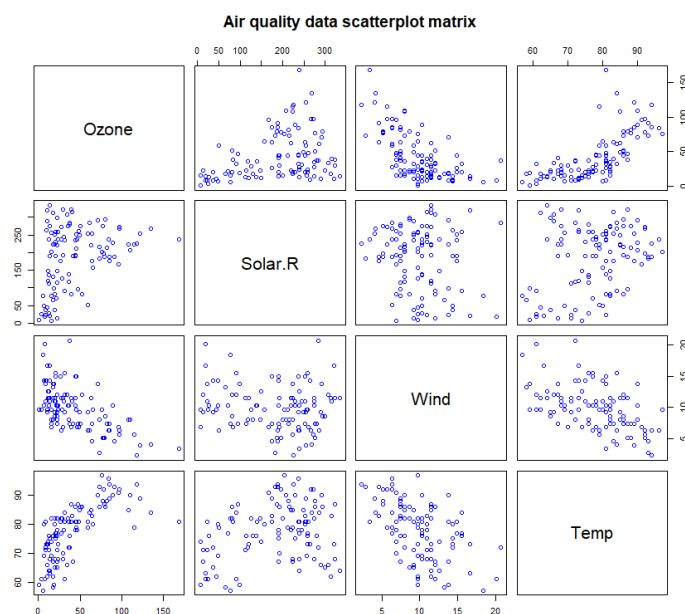
The second visualization is a boxplot that represents the distribution of ozone levels values grouped by every month of the observation (Fig. 8). The orange boxes are the intervals of values registered for every month presented. The bigger discrepancies between the ozone levels were registered in july and august and the smallest discrepancy was registered in June. For those orange boxes, the lower line represents the values under the 1st quartile and the top line the values under the 3rd quartile. The thick black line from every orange box is the median value.

The dashed lines that goes of every bottom and top of the orange boxes are the lowest and highest values which are still not outliers and the small circles are the outliers from every month.



(Fig. 8 “Month comparison for the ozone levels”)

The third visualization is a scatterplot of the variables ozone, solar radiation, wind and temperature (Fig. 9). The scatterplot confirms the relationships between the variables. The plot of ozone and temperature (top right) shows a strong positive correlation, suggesting that hot days are associated with higher pollution. The plot of ozone and wind (top middle) shows a negative trend, indicating that stronger winds help disperse ozone. In help for this scatterplot come also a correlation matrix that confirm with the help of numbers what was presented above (Fig. 10)



(Fig. 9 “Air quality scatterplot”)

	Ozone	Solar.R	wind	Temp
Ozone	1.0000000	0.3483417	-0.6124966	0.6985414
Solar.R	0.3483417	1.0000000	-0.1271835	0.2940876
wind	-0.6124966	-0.1271835	1.0000000	-0.4971897
Temp	0.6985414	0.2940876	-0.4971897	1.0000000

(Fig. 10 "Correlation matrix")

After all the visualizations were created, the next step was to create some hypothesis tests in order to validate the differences between variables of the data set. The tests used in this project are:

- a) ANOVA test – It was tested the null hypothesis that the ozone levels are the same across all months (Fig. 11). The p-value 1,78e-05 is less then the standard threshold of 0,05, so the result is highly statistically significant. The null hypothesis is rejected and the alternative one is that at least one month has a different ozone level.

```
> print("ANOVA results:")
[1] "ANOVA results:"
> anova_model <- aov(ozone ~ Month, data = aq_clean)
> summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Month	4	27383	6846	7.686	1.78e-05 ***
Residuals	106	94419	891		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Fig. 11 "ANOVA test")

- b) Test for distribution of data – it was used to validate ANOVA assumptions, testing the distribution of ozone (Fig. 12). The p-value 2,846e-08, far below the significance level of 0,05 confirms the data is not normally distributed.

```
> #Testing the distribution of the data
> print("Distribution of the data test results:")
[1] "Distribution of the data test results:"
> shapiro.test(aq_clean$ozone)
```

shapiro-wilk normality test

```
data:  aq_clean$ozone
W = 0.87355, p-value = 2.846e-08
```

(Fig. 12 "Test for distribution of data")

- c) Nonparametric test – it is used on the data which is not normally distributed. The p-value 2,742e-05, another very small value, less than the 0,05 threshold, confirms once more the ANOVA test, so there is a significant difference air quality across months (Fig. 13).

```
> #Nonparametric test
> print("Nonparametric test results:")
[1] "Nonparametric test results:"
> kruskal.test(Ozone ~ Month, data = aq_clean)

Kruskal-wallis rank sum test

data: Ozone by Month
Kruskal-wallis chi-squared = 26.309, df = 4, p-value = 2.742e-05
```

(Fig. 13 "Nonparametric test")

- d) Test for proportions – It was compared the proportion of the days with a high temperature, those being may and august, both with over 80 degrees Fahrenheit and it was created a contingency table to represent the difference between the days with the mentioned temperature levels (Fig. 14). It can be observed that there exists a significant difference, august (14 days) having a much higher proportion of hot days than may (1 day).

```
> #Test for proportions
> may_aug <- subset(aq_clean, Month %in% c(5, 8))
> may_aug$Month <- droplevels(may_aug$Month)
>
> #Creating a contingency table to compare the month and high temp variables
> prop_table <- table(may_aug$Month, may_aug$Temp > 80)
> print("Table for proportions:")
[1] "Table for proportions:"
> print(prop_table)
```

	FALSE	TRUE
5	23	1
8	9	14

(Fig. 14 "Test for proportions and contingency table")

- e) T-test – It was performed an unpaired t-test to compare the average temperature of the hottest months as presented in the last point, may and august (Fig. 15). The result showed a statistically significant difference between means of the two months, the p-value being 4,779e-11, far less then the standard significance of 0,05. In conclusion, the null hypothesis, that says that the means are equal, is rejected, and the alternative is accepted, the means of the temperature of the two months are different.

```
> #Runing t-test
> t.test(Temp ~ Month, data = may_aug)

welch Two Sample t-test

data: Temp by Month
t = -8.6219, df = 44.51, p-value = 4.779e-11
alternative hypothesis: true difference in means between group 5 and group 8 is not equal to 0
95 percent confidence interval:
 -21.26525 -13.20938
sample estimates:
mean in group 5 mean in group 8
 66.45833      83.69565
```

(Fig. 15 "T-test")

The last point of this project was to create a multiple linear regression predictive model (Fig. 16) that predicts the ozone level for any given day in the data set. There were used three independent variables in the model which are the solar radiation, wind speed and temperature. What the model does is to calculate a specific weight for each variable. The algorithm calculates the optimal coefficients that minimize the sum of the squared errors between the predicted and the actual ozone levels.

```
#Predicting the ozone level using the other 3 variables
model <- lm(Ozone ~ Solar.R + wind + Temp, data = aq_clean)

summary(model)

#Predicting the ozone level for a random day
new_data <- data.frame(Solar.R = 190, wind = 7.4, Temp = 80)

prediction <- predict(model, newdata = new_data)
print(paste("Predicted ozone level:", round(prediction, 2)))

par(mfrow = c(2, 2))
plot(model)
par(mfrow = c(1, 1))
```

(Fig. 16 “Predictive model code”)

The “predict()” function was used to estimate the ozone level for a chosen set of meteorological conditions (solar radiation = 190; wind speed = 7,4; temperature = 80). After that function was used to calculate the prediction (Fig. 17), some diagnostic plots were generated (Fig. 18).

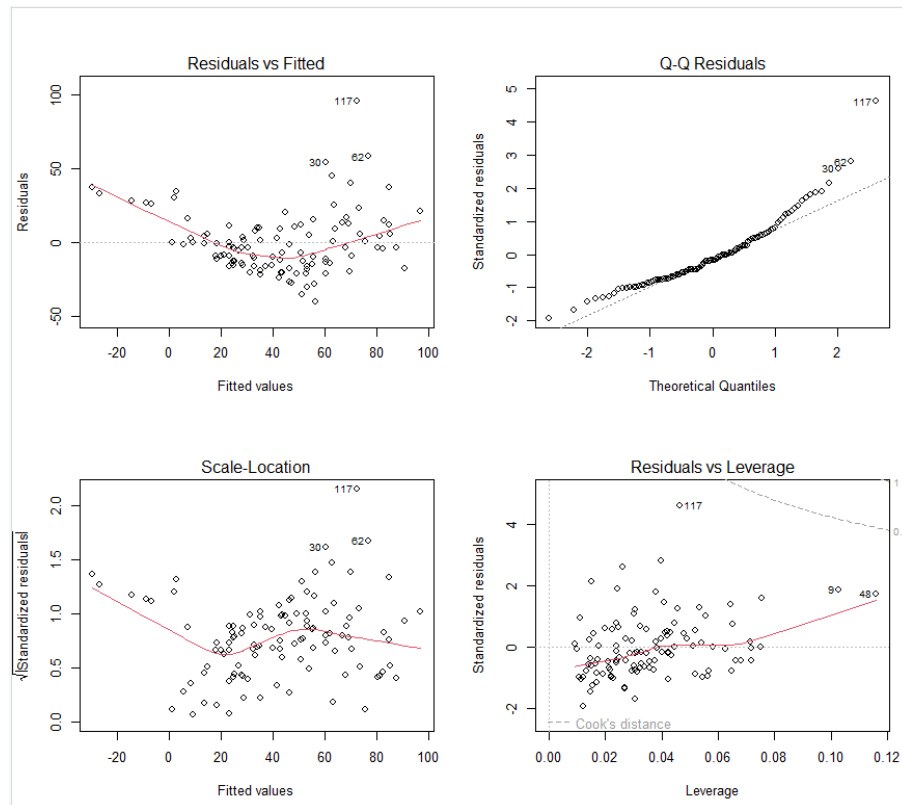
```
call:
lm(formula = ozone ~ Solar.R + wind + Temp, data = aq_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-40.485 -14.219  -3.551   10.097   95.619

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -64.34208    23.05472  -2.791  0.00623 **
Solar.R       0.05982     0.02319   2.580  0.01124 *
wind        -3.33359     0.65441  -5.094 1.52e-06 ***
Temp         1.65209     0.25353   6.516 2.42e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21.18 on 107 degrees of freedom
Multiple R-squared:  0.6059,    Adjusted R-squared:  0.5948
F-statistic: 54.83 on 3 and 107 DF,  p-value: < 2.2e-16
[1] "Predicted ozone level: 54.52"
```

(Fig. 17 “Predictive model output”)



(Fig. 18 “Predictive model plots”)

The predictive model generated significant results (statistics $F = 54,83$ and $p\text{-value} = 2,2e-16$, less than 0,05). The model explains approx. 60,6% of the variance of the ozone level ($R^2 = 0,6059$).

The temperature had the strongest effect (coefficient = 1,65 and $p\text{-value} = 0,01124$) which means that for every 1 degree Fahrenheit increase, ozone levels increases too with 1,65 units.

The wind speed had a significant negative effect (coefficient = -3,33 and $p\text{-value} = 1,52e-06$), meaning that a decrease with 1 mph of wind is decreasing the ozone layer level with 3,33.

The solar radiation had a smaller but significant positive effect (coefficient = 0,06 and $p\text{-value} = 2,42e-09$), so a increase of 1 unit of solar radiation, is increasing the ozone layer with 0,06 units.

The diagnostic plots confirm the model’s validity with some limitations. The Residuals vs Fitted plot displays a slight curvature, presenting that the relationship between predictors and ozone may be non linear. The Normal Q-Q Residual plot shows that residuals are, in a large part, normally distributed, though deviations in the upper tail reflect the model's difficulty in predicting extreme outlier values. The Scale-Location plot reveals a slight upward trend in the red line and a wider spread of points on the right side. This means that the model predictions become less precise when ozone level increase. On the other hand, the Residuals vs Leverage plot confirms that no influential outliers are distorting the model parameters.

4. Conclusion

Based on the statistical analysis performed on the “airquality” dataset, it can be concluded that the ozone levels are not consistent. The hypothesis testing, made on this project, confirmed with high statistical significance that air quality varies depending on the season. The data showed that ozone pollution peaks during the months with a higher temperature, July and August, which had higher variability and big outliers compared to the cooler months as May and September.

The analysis identified strong relationships between weather conditions and pollution levels. The correlation matrix and scatterplots revealed that temperature is the most significant driver of poor air quality. As temperatures rise, ozone levels increase. In contrast, wind speed acts as a cleaning factor, showing a strong negative correlation where higher wind speeds help disperse the pollution. Solar radiation also contributes to higher ozone levels, though its impact is smaller compared to temperature and wind.

Finally, the Multiple Linear Regression model developed in this project proved to be a useful tool for prediction, explaining approximately 60,6% of the variation in ozone levels. While the model successfully captured the general trends, the diagnostic plots presented some limitations. Since the ozone data contains big outliers, the model becomes less precise when predicting high pollution days. However, the overall results confirm that meteorological data can be effectively used to estimate air quality risks, provided that seasonal patterns and extreme weather events are significant factors.